# MRI Full Course Formula Sheet

Compact formulas for exam memorization - two lectures per page

## Week 1 - NMR Basics

### Spin

| | | |
|---|---|---|
| Angular momentum | $p = mv$, $J = r \times p$ | Rotating charge. |
| Magnetic moment | $\mu = \gamma J$ | γ gyromagnetic ratio. |
| Quantization | $J_z = \hbar m$, $m = I, I-1, \ldots, -I$ | Spin projection. |
| Spin 1/2 | $I = 1/2 \Rightarrow m = \pm 1/2$ | $^{1H}$ has two states. |
| Moment projection | $\mu_z = \gamma\hbar m = \pm(1/2)\gamma\hbar$ | Along $B_{0.}$ |

### Energy And Resonance

| | | |
|---|---|---|
| Energy | $E_m = -\mu_{zB0} = -\gamma\hbar B_{0m}$ | Parallel lower for γ>0. |
| Gap | $\Delta E = \gamma\hbar B_0 = \hbar\omega_0$ | Energy splitting. |
| Boltzmann | $n_{down/nup} = e^{-}\Delta E/k_{BT}$ | Population ratio. |
| Polarization | $\Delta n/n \approx \Delta E/(2k_{BT)}$ | For ΔE << kBT. |
| Net M | $M_0 = \Sigma\mu = \Delta n\ \mu_{zez}$ | Longitudinal M. |
| Scaling | $M_0 \propto \rho_{HB0/T}$ | Proton density matters. |

### Dynamics

| | | |
|---|---|---|
| Torque | $d\mu/dt = \gamma(\mu \times B)$ | Precession. |
| Macroscopic | $dM/dt = \gamma(M \times B)$ | Same law for M. |
| RF | $B_{1(t)}=B_{1cos(}\omega_{0t)ex}=B_{L+BR}$ | Excitation field. |
| Rotating frame | $(dM/dt)_{rot}=\gamma M\times B-\omega_0\times M=\gamma M\times B_{eff}$ | Use effective field. |
| Effective field | $B_{eff}=(B_{0-}\omega/\gamma)e_z'+B_1,rote_x'$ | On resonance mostly B1. |
| Flip angle | $\alpha = \gamma\int B_1,rot(t)dt$ | RF pulse area. |

### Relaxation

| | | |
|---|---|---|
| T2* | $1/T_{2*} = 1/T_2 + \gamma\delta B$ | Field inhomogeneity. |
| Bloch x | $dM_{x/dt}=\gamma(M_{yBz-MzBy)}\ -M_{x/T2}$ | Transverse loss. |
| Bloch y | $dM_{y/dt}=\gamma(M_{zBx-MxBz)}\ -M_{y/T2}$ | Transverse loss. |
| Bloch z | $dM_{z/dt}=\gamma(M_{xBy-MyBx)}\ -(M_{z-M0)/T1}$ | Recovery. |

## Week 2 - Image Formation

### Encoding

| | | |
|---|---|---|
| Gradient | $G_i=\partial B_{z/}\partial i$ | i=x,y,z. |
| Linear field | $\Delta B_{z(x)}=G_{xx}$ | Constant slope. |
| Frequency | $\omega(r)=\gamma(B_{0+G}\ r)$ | Position dependent. |
| Readout | $\omega(x)=\gamma G_{xx}$ | Frequency encoding. |
| Phase | $\varphi(y)=\gamma G_{yyTy}$ | Phase encoding. |
| K-space | $k(t)=\gamma\int G(t)dt$ | Constant G: k=γGt. |

### Signal And FT

| | | |
|---|---|---|
| Signal | $s(k_x,k_{y)}\propto\Sigma\Sigma\rho(x_i,y_{j)e}{}^{j(k}{}_{xxi+kyyj)}$ | FT of object. |
| Image | $\rho(x,y)\propto\Sigma\Sigma s(k_{xp},k_{yq)e}{}^{-j(k}{}_{xpx+kyqy)}$ | Inverse FT. |
| K spacing | $\Delta k=2\pi/FOV$ | Nyquist. |
| Pixel | $\Delta x=FOV_{x/Nx}$ | Same in y. |
| Max k | $k_{max}=\pi/\Delta x=\pi N_{x/FOVx}$ | Resolution. |
| Readout BW | $BW_x=\gamma G_{xFOVx}$ | Angular freq convention. |
| Dwell | $\Delta t=2\pi/BW_x$, $T_x=N_x\Delta t$ | Sampling. |

### Slice And Contrast

| | | |
|---|---|---|
| Gradient moment | $k_i=\gamma\int G_{i(t)dt}$ | Gradient area. |
| Slice thickness | $\Delta z=BW_{RF/(}\gamma G_{z)}$ | RF bandwidth. |
| RF selectivity | $R=BW_{RFTRF}$ | Longer pulse narrower BW. |
| PD contrast | $C\propto\rho_{A-}\rho_B$ | Long TR. |
| Partial sat. | $C\propto\rho_{A(1-e}{}^{-TR/T}{}_{1A)-}\rho_{B(1-e}{}^{-TR/T}{}_{1B)}$ | T1 weighting. |
| Scan time | $T_{scan}\approx N_{phaseTR}$ | One ky line per TR. |

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## Week 3 - Fast And Parallel Imaging

### Speed, SNR, Resolution

| | | |
|---|---|---|
| Voxel | $\Delta V=\Delta x\Delta y\Delta z$ | Voxel volume. |
| SNR scaling | $SNR \propto \Delta V\sqrt{T}_{scan/}\sqrt{BW}$ | Core tradeoff. |
| Averages | $SNR \propto \sqrt{NSA}$ | Repeated scans. |
| Acceleration | $T_{scan}, R\approx T_{scan/R}$ | Undersampling. |
| Aliased FOV | $FOV_{alias}=FOV/R$ | Phase undersampling. |

### Coil Encoding

| | | |
|---|---|---|
| Coil signal | $s\_\gamma(k)=\int\rho(x)c\_\gamma(x)e^{ikxdx}$ | Sensitivity cγ. |
| Matrix model | $s = E\rho + \eta$ | Encoding plus noise. |
| Decode | $i = Fs$ | Reconstruction matrix. |
| SENSE | $\hat{\rho}=(E^{H}\Psi^{-1E)-1EH}\Psi^{-1s}$ | Optimum SNR inverse. |
| Regularized | $\hat{\rho}=(E^{H}\Psi^{-1E+}\lambda I)^{-1EH}\Psi^{-1s}$ | Stabilizes ill-conditioning. |

### Parallel Limits

| | | |
|---|---|---|
| SENSE SNR | $SNR_{SENSE}=SNR_{full/(}\sqrt{R}\cdot g(x))$ | g-factor penalty. |
| g-factor | $g(x)=\sqrt{[(E^{HE)}{}_{ii((E}{}^{HE)-1)}{}_{ii}]} \geq 1$ | Noise amplification. |
| Good coils | Distinct $c\_\gamma(x) \Rightarrow$ low g | Separates aliased voxels. |
| Failure | $R > N_{coils}$ or poor sensitivities | Underdetermined/ill-conditioned. |

## Week 4 - Image Contrast

### Relaxation Contrast

| | | |
|---|---|---|
| T1 recovery | $M_{z(t)}=M_{0+}[M_{z(0)-M0}]e^{-t/T}{}_{1}$ | Longitudinal. |
| After 90° | $M_{z(t)}=M_{0(1-e}{}^{-t/T}{}_{1)}$ | Mz(0)=0. |
| T2 decay | $M_{xy(t)}=M_{xy(0)e}{}^{-t/T}{}_{2}$ | Spin-spin. |
| T2* decay | $M_{xy(t)}=M_{xy(0)e}{}^{-t/T}{}_{2*}$ | GRE/FID. |

### Common Signals

| | | |
|---|---|---|
| Spin echo | $S_{SE}\propto\rho(1-e^{-TR/T}{}_{1)e}{}^{-TE/T}{}_{2}$ | T1 by TR, T2 by TE. |
| GRE | $S_{GRE}\propto\rho\sin\alpha\,(1-E_{1)/(1-E1cos}\alpha)e^{-TE/T}{}_{2*}$ | $E1=e^{-TR/T1.}$ |
| Inversion | $M_{z(TI)}=M_{0(1-2e}{}^{-TI/T}{}_{1)}$ | TR >> T1. |
| Null time | $TI_{null}=T_{1ln2}$ | Suppress tissue. |
| Ernst | $\alpha_{E}=\arccos(e^{-TR/T}{}_{1)}$ | Max steady-state signal. |

### Weighting Rules

| | | |
|---|---|---|
| PD | TR long, TE short | Minimize relaxation weighting. |
| T1 | TR short, TE short | T1 recovery dominates. |
| T2 | TR long, TE long | T2 decay dominates. |
| T2* | GRE with TE long | Sensitive to dephasing. |

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## Week 5 - SNR And Hardware

### Signal And Noise

| | | |
|---|---|---|
| Signal voltage | $U_{sig(x)}=\omega M_{xy(x)C(x)}\Delta V$ | Receive sensitivity C. |
| Noise power | $P=\sigma\|E\|^2\Delta V$ | Sample losses. |
| Noise variance | $\Psi=4k_{BTBW}$ R | Johnson-Nyquist. |
| SNR | $SNR=U_{sig/Unoise}$ | Image quality. |
| SNR scaling | SNR∝ΔV√NSA/√BW | Resolution/speed tradeoff. |

### Optimization

| | | |
|---|---|---|
| Increase signal | ↑C, ↑$M_{xy}$, ↑ω, ↑ΔV | More voltage. |
| Reduce noise | ↓BW, ↓T, ↓$R_{coil}$ | Less thermal noise. |
| Field scaling | $M_0 \propto B_0$, $\omega \propto B_0$ | $U_{sig}$ roughly grows strongly with B0. |
| Surface coil | C(x) high near coil | High local SNR. |

### Resolution Limits

| | | |
|---|---|---|
| Fourier | $\Delta x \approx \pi/k_{max}$ | Sampling aperture. |
| Relaxation blur | $k^* = \gamma GT_{2*}$ | Finite T2* filters k-space. |
| Diffusion blur | $x^2 \approx 6DT_{acq}$ | Long readouts blur. |
| SoS combine | $I_{SoS}=\sqrt{\Sigma_c\|I_c\|^2}$ | Magnitude coil combine. |

## Week 6 - Flow Imaging

### Motion Phase

| | | |
|---|---|---|
| Trajectory | $r(t)=r_{0+v(t-t0)+a(t-t0)}^{2/2+...}$ | Moving spin. |
| Phase | $\varphi=\gamma\int G(t)r(t)dt$ | Gradient phase. |
| Moments | $M_n=\int G(t)(t-t_{0)}^{ndt}$ | nth gradient moment. |
| Expansion | $\varphi=\gamma[r_{0M0+vM1+aM2/2+...}]$ | Position/velocity/acceleration. |
| Bipolar | $M_0=0$, $\varphi \approx \gamma vM_1$ | Velocity encoding. |

### Phase Contrast

| | | |
|---|---|---|
| Encoding | $VENC=\pi/(\gamma M_{1)}$ | Phase reaches ±π. |
| Velocity | v=(Δφ/π)VENC | Phase difference map. |
| Aliasing | \|v\|>VENC ⇒ phase wraps | Set VENC high enough. |
| Flow rate | $Q=\Sigma\ v_i\Delta A_i$ | Through-plane flow. |
| Velocity distribution | $s(x,k_{v)}=\Sigma\rho(x,v)e^{jk}_{vv}$ | Generalized velocity encoding. |

### Hemodynamics

| | | |
|---|---|---|
| Poiseuille | $Q=\Delta P/R$, $R \propto 1/d^4$ | Diameter dominates. |
| Reynolds | Re=ρvd/μ | Laminar if Re<~2000. |
| Shear stress | SS=-μ ∂v/∂r | Wall shear. |
| Contrast agent | $1/T_1$,app=$1/T_{1+R1c}$ | Relaxivity model. |

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## Week 7 - Motion And Artifacts

### Motion Encoding

| | | |
|---|---|---|
| Phase | $\varphi=\gamma\int G(t)x(t)dt$ | All motion artifacts start here. |
| Motion model | $x(t)=x_{0+v(t-t0)+a(t-t0)}^{2/2+...}$ | Taylor expansion. |
| Moment nulling | $M_0=0$ removes position phase; $M_1=0$ compensates velocity | Gradient design. |
| Velocity phase | $\varphi_v=\gamma v M_1$ | Residual if M1 nonzero. |
| Velocity spread | signal $\propto$ sinc($\beta\gamma A\Delta t\ \Delta y/2$) | Intravoxel dephasing. |

### Artifact Rules

| | | |
|---|---|---|
| Intra-TR | motion during readout/echo ⇒ phase errors and signal loss | Flow, pulsation. |
| Inter-TR | motion between ky lines ⇒ ghosting/blurring | Respiration/patient. |
| Periodic ghosts | ghost spacing in phase direction $\propto$ $1/(f_{motionTR)}$ | Regular motion. |
| Random motion | random ky phase ⇒ diffuse blur | Irregular motion. |

### Correction

| | | |
|---|---|---|
| Rigid transform | $R(r,t)=A(t)r+d(t)$ | Rotation plus translation. |
| Prospective gradients | $G'(t)=A(t)^{-TG(t)}$ | Follow anatomy. |
| Phase correction | $\varphi'(t)=\varphi(t)-\gamma G(t)\cdot d(t)$ | Translation compensation. |

## Week 8 - fMRI And Diffusion

### BOLD/fMRI

| | | |
|---|---|---|
| T2* | $1/T_{2*}=1/T_{2+}\gamma\Delta B$ | Static dephasing. |
| Susceptibility | $\Delta B\approx\Delta\chi B_0$ | Blood oxygenation effect. |
| GRE BOLD | $S(TE)=S_{0e}^{-TE/T}{}_{2*}$ | TE near T2*. |
| Small change | $\Delta S/S\approx-TE\cdot\Delta R_{2*}$ | R2*=1/T2*. |

### Diffusion Physics

| | | |
|---|---|---|
| Fick | $j=-D\nabla c(r,t)$ | Diffusion flux. |
| Diffusion eq. | $\partial c/\partial t=D\nabla^{2c}$ | Conservation plus Fick. |
| RMS displacement | $R_{rms}=\surd(6D\Delta)$ | 3D free diffusion. |
| Einstein | $D\approx v^2\tau/6$ | Random walk. |

### DWI/DTI

| | | |
|---|---|---|
| DWI signal | $S(TE,b)=S_{0e}^{-TE/T}{}_{2e}^{-bD}$ | Scalar diffusion. |
| b-value | $b=\gamma^{2G2}\delta^{2(}\Delta-\delta/3)$ | PGSE sensitivity. |
| Tensor | $\ln(S/S_{0)}=-b\ g^{TDg}$ | Direction g. |
| MD | $MD=(\lambda_{1+}\lambda_{2+}\lambda_{3)/3}$ | Mean diffusivity. |
| FA | $FA=\surd(3/2)\ \surd\Sigma(\lambda_{i-MD)}^{2/}\surd\Sigma\lambda_i^2$ | Anisotropy. |

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## Week 9 - Advanced Imaging

### General Encoding

| | | |
|---|---|---|
| Data model | $d=E\rho+\eta$ | General MRI inverse problem. |
| Cartesian FT | E=Fourier sampling operator | Standard MRI. |
| Parallel | E includes coil sensitivities c_γ(r) | SENSE/arrays. |
| Optimal inverse | $\hat{\rho}=(E^{H}\Psi^{-1E)-1EH}\Psi^{-1d}$ | If well-conditioned. |
| Regularized | $\hat{\rho}=(E^{H}\Psi^{-1E+}\lambda I)^{-1EH}\Psi^{-1d}$ | For high R. |

### Compressed Sensing

| | | |
|---|---|---|
| Sparsity | $x=\Phi\rho$ has many small/zero coefficients | Sparse transform. |
| Undersampled data | $d\_\Omega=P\_\Omega F\rho$ | Sample subset Ω. |
| CS recon | $\min\_\rho\ \|\|\Phi\rho\|\|_1$ s.t. $\|\|E\rho-d\|\|_2\le\varepsilon$ | Sparse recovery. |
| TV | $\Phi\rho=\nabla\rho \Rightarrow$ total variation penalty | Piecewise smooth images. |
| Sampling | random/incoherent k-space undersampling | Artifacts become noise-like. |

### Limits

| | | |
|---|---|---|
| PSF | $PSF(r)=FT^{-1}$\{sampling pattern\} | Undersampling artifact shape. |
| Acceleration | higher R needs sparsity + SNR + calibration | No free lunch. |
| Low-rank/dynamics | suppp small or temporal basis small | Advanced priors. |

## Week 10 - Spectroscopy I

### Chemical Shift

| | | |
|---|---|---|
| Larmor | $\nu=-\gamma B_0$ [Hz] | Sign by convention. |
| Shift | $\delta=(\nu-\nu_{ref)/}\nu_{ref}\cdot 10^{6}$ ppm | Field-independent ppm. |
| Hz separation | $\Delta\nu=\Delta\delta\cdot\nu_{ref}\cdot 10^{-6}$ | Grows with B0. |
| 1 ppm | 1 ppm = $\nu_{ref}\cdot 10^{-6}$ Hz | 64 Hz at 1.5T for 1H. |

### Spectrum

| | | |
|---|---|---|
| FID | $s(t)=A\ e^{-t/T}{}_{2^*e}{}^{i2}\pi\nu_{At}$ | Single component. |
| Spectrum | $S(\nu)=FT\{s(t)\}$ | Frequency-domain signal. |
| Lorentzian | $\mathrm{Re}\ S(\nu)\propto T_{2^*} / [1+(2\pi T_{2^*(}\nu-\nu_{A))}{}^{2}]$ | Line shape. |
| Linewidth | $FWHM=1/(\pi T_{2^*)})$ | Shorter T2* broader. |
| Mixture | $s(t)=\Sigma_{cwce}{}^{-t/T}{}_{2,}c^*e^{i2}\pi\nu_{ct}$ | Superposition. |

### Sampling And J

| | | |
|---|---|---|
| Acquisition | $T_{acq}=Ndt$ | N samples. |
| Bandwidth | $BW=1/dt$ | Hz. |
| Resolution | $d\nu=1/T_{acq}$ | Peak spacing. |
| Resolve peaks | $d\nu\le\Delta\nu/2$ | Rule of thumb. |
| J multiplet | N coupled spin-I nuclei ⇒ 2NI+1 lines | Binomial intensities. |
| J echo | $TE=n/J$ | Odd n flips doublet sign. |
| Editing | $TE=1/J$ | Difference editing. |

## Week 11 - Spectroscopy II

### Suppression And Inversion

| | | |
|---|---|---|
| Water/fat | $\Delta\delta \approx 3.5$ ppm | Water 4.7, fat 1.2 ppm. |
| Phase accrual | $\Delta\varphi = 2\pi\Delta\nu\tau$ | Chemical shift phase. |
| Binomial spacing | $\tau \approx 1/(2\Delta\nu)$ or $1/(4\Delta\nu)$ | Depends on target phase. |
| Inversion recovery | $M_{z(TI)}=M_{0(1-2e}{}^{-TI/T}{}_{1)}$ | After 180°. |
| Null | $TI_{null}=T_{1\ln 2}$ | Water/fat suppression. |

### Localization

| | | |
|---|---|---|
| Surface coil | $S(r)=B_{1(r)}\sin[\gamma B_{1(r)t}]$ | Sensitivity + flip angle. |
| Adiabatic | $\lvert d\theta/dt\rvert \ll \gamma\lvert B_{eff}\rvert$ | Robust inversion/excitation. |
| PRESS/STEAM | Echo at TE; stimulated echo stores Mz during TM | Voxel selection. |
| Phase cycling | linear combinations isolate voxel term | Remove unwanted echoes. |

### CSI And High Field

| | | |
|---|---|---|
| Shift artifact | $\Delta x=\Delta\nu_{CS/(}\gamma G)$ | Chemical shift displacement. |
| Fractional shift | $\Delta x/\Delta x_{RF}=\Delta\nu_{CS/BWRF}$ | Use large RF BW. |
| CSI signal | $s(k_x,k_y,t)=\iiint\rho(x,y,\Delta\nu)e^{ik}{}_{xxe}{}^{ik}{}_{yye}{}^{i2}\pi\Delta\nu t\,dx\,dy\,d\Delta\nu$ | 2 spatial + spectral. |
| CSI recon | $S(x,y,\Delta\nu)=FT_{tFTkxFTky}\{s\}$ | 3D transform. |
| CSI scan time | $T_{scan}=N_{xNyTR}\cdot NSA$ | Slow phase encoding. |
| High-field SNR | $U_{sig}\sim B_0^2$, $U_{noise}\sim B_0$, $SNR\sim B_0$ | Approximate. |
| SAR | $P\propto\sigma E^2\propto\sigma\gamma^{2B}{}_0{}^{2B}{}_1{}^2$ | High-field cost. |